



EXAM PAPERS PRACTICE

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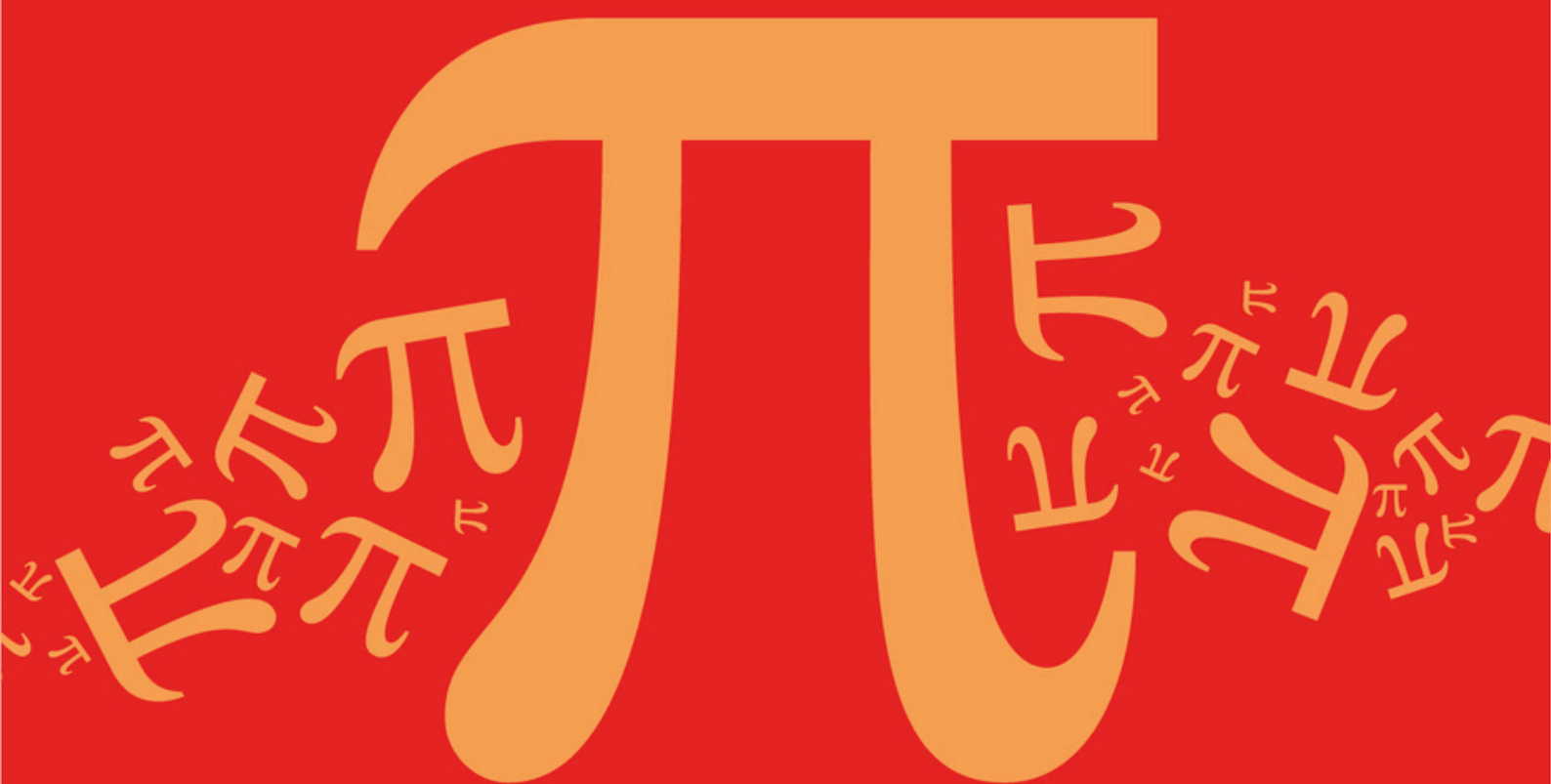
Practice questions created by actual examiners and assessment experts

Detailed mark scheme

Suitable for all boards

Designed to test your ability and thoroughly prepare you

AQA A Level Further Mathematics (7367)



Topic

D: Further Algebra and Functions

Sub topic

D12: Graphs of Rational Functions

Topic Questions

AQA A Level Further Mathematics (7367)

Question Paper

Topic

D: Further Algebra and Functions

Sub topic

D12: Graphs of Rational Functions

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**To be used by all students and teachers for
AQA A Level Further Mathematics (7367)
for exams from 2025 onwards.**

Students and teachers of other boards may also find this useful

Name : _____

Date : _____

Time : _____

Total Marks Available : _____

Total Marks Received : _____

Q1.

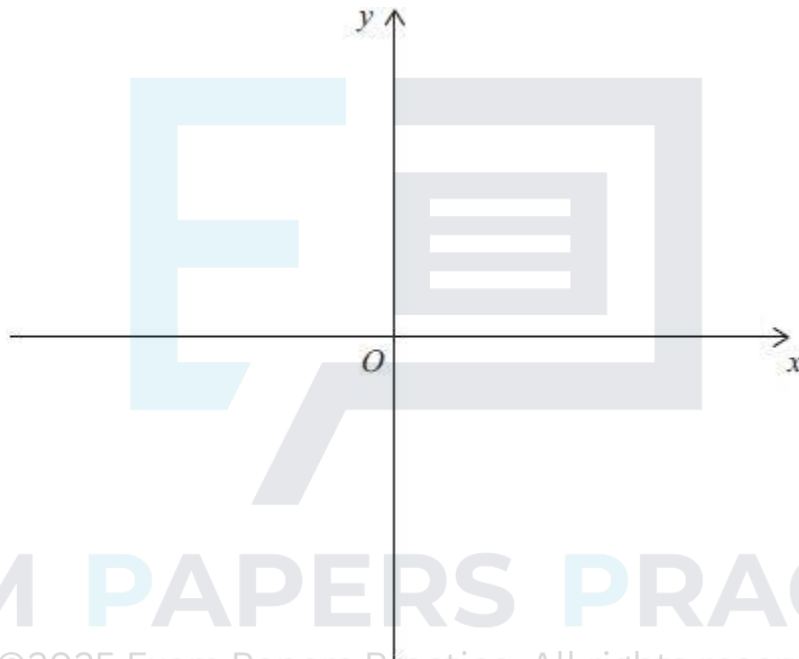
The graph of the rational function $y = f(x)$ intersects the x -axis exactly once at $(-3, 0)$

The graph has exactly two asymptotes, $y = 2$ and $x = -1$

(a) Find $f(x)$

(2)

(b) Sketch the graph of the function.



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(3)

(c) Find the range of values of x for which $f(x) \leq 5$

(4)

(Total 9 marks)

Q2.

A curve has equation

$$y = \frac{x}{x-1}$$

(a) Find the equations of the asymptotes of this curve.

(2)

(b) Given that the line $y = -4x + c$ intersects the curve, show that the x -coordinates of the points of intersection must satisfy the equation

$$4x^2 - (c + 3)x + c = 0$$

(3)

(c) It is given that the line $y = -4x + c$ is a tangent to the curve.

- (i) Find the two possible values of c .

(No credit will be given for methods involving differentiation.)

(3)

- (ii) For each of the two values found in part (c)(i), find the coordinates of the point where the line touches the curve.

(4)

(Total 12 marks)

Q3.

A curve has equation $y = \frac{1}{x^2 - 4}$.

- (a) (i) Write down the equations of the three asymptotes of the curve.

(3)

- (ii) Sketch the curve, showing the coordinates of any points of intersection with the coordinate axes.

(4)

- (b) Hence, or otherwise, solve the inequality

$$\frac{1}{x^2 - 4} < -2$$

(3)

(Total 10 marks)

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Q4.

- (a) (i) Write down the equations of the two asymptotes of the curve $y = \frac{1}{x - 3}$.

(2)

- (ii) Sketch the curve $y = \frac{1}{x - 3}$, showing the coordinates of any points of intersection with the coordinate axes.

(2)

- (iii) On the same axes, again showing the coordinates of any points of intersection with the coordinate axes, sketch the line $y = 2x - 5$.

(1)

- (b) (i) Solve the equation

$$\frac{1}{x - 3} = 2x - 5$$

(3)

- (ii) Find the solution of the inequality

$$\frac{1}{x-3} < 2x - 5$$

(2)

(Total 10 marks)

Q5.

A curve has equation

$$y = \frac{3x-1}{x+2}$$

- (a) Write down the equations of the two asymptotes to the curve. (2)
- (b) Sketch the curve, indicating the coordinates of the points where the curve intersects the coordinate axes. (5)
- (c) Hence, or otherwise, solve the inequality

$$0 < \frac{3x-1}{x+2} < 3$$

(2)

(Total 9 marks)

Q6.

A curve has equation

$$y = \frac{6x}{x-1}$$

- (a) Write down the equations of the two asymptotes to the curve. (2)
- (b) Sketch the curve and the two asymptotes. (4)
- (c) Solve the inequality

$$\frac{6x}{x-1} < 3$$

(4)

(Total 10 marks)